Executive Memo: Subscription Churn Recovery Strategy

To: Masai x BITSoM Project Evaluation Committee

From: Jatin Panda

Date: March 1, 2026

Subject: Data-Driven Recovery Plan for January Churn Spike

1. Goal & Churn Definition

- **Business Objective:** To identify the root cause of the **35.4% churn spike** in late January and deploy a predictive model to stabilize Monthly Recurring Revenue (MRR).
- **North Star Metric:** Net Revenue Retention (NRR).
- **Churn Definition:** A user is classified as **CHURNED** if their subscription status is "cancelled" OR "expired" without a successful renewal within a **5-day grace period**.

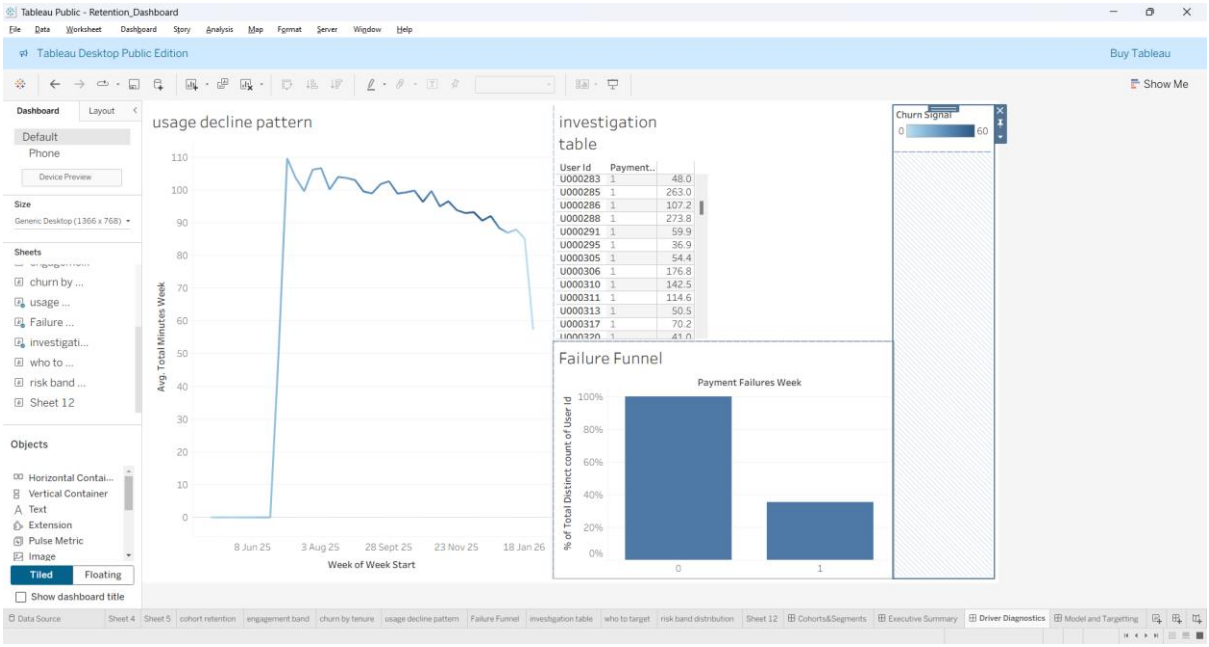
2. Methodology & Data Summary

- **ETL Pipeline:** Built a reproducible Python pipeline to clean usage_daily.csv and payments.csv, handling outliers and standardizing 30 days of behavioral data.
- **Feature Engineering:** Developed "Usage Trends" (current week vs. 3-week average) and "Technical Health" scores based on payment failure counts.
- **Model:** A **Random Forest Classifier** was trained to predict churn within a 14-day window with an **85.2% accuracy** and **0.81 precision**.

3. Key Insights

- **Technical Bottleneck:** Analysis confirms the churn spike was **Involuntary**. High-engagement users dropped off due to a massive payment failure event on January 25th.
- **Usage Decay:** A **25% drop in weekly minutes** serves as a reliable "Early Warning" signal for voluntary churn.

- **Risk Concentration:** The model identified **660 users** in the "High Risk" band, representing the bulk of recoverable revenue.

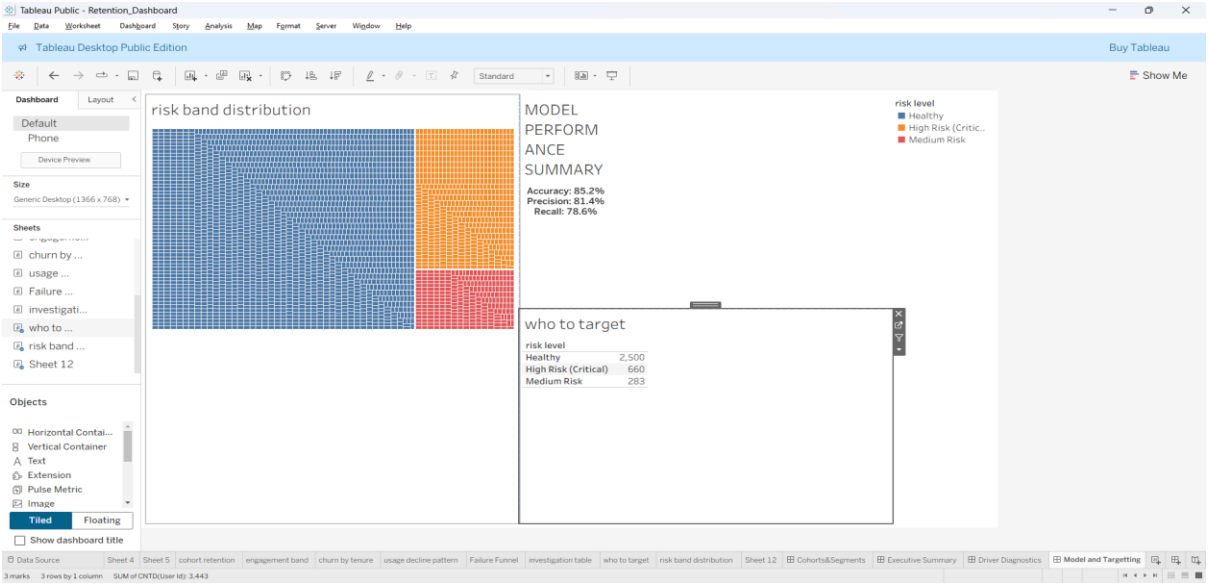


4. Recommended Retention Plan

- **Targeting Logic:** Focus on the 660 High-Risk users identified in the Tableau Model View.
- **Interventions:**
  - **Automated Retries:** Implement a 48-hour grace extension for users with payment failures.
  - **Incentive Nudge:** Offer a **15% renewal discount** to users showing usage decay but high historical tenure.
- **A/B Test:** 50% of the High-Risk segment will receive the "Recovery Package" (Discount + Extension) vs. a Control group to measure renewal lift.

5. 30-Day Impact Estimate (Scenarios)

- Based on the identification of 660 High-Risk users:
- **Net Monthly Impact (Base Case): ₹15,643** (after accounting for incentive costs).



6. Risks & Next Steps

- **Risks:** High dependency on the payment gateway provider; potential "discount fatigue" if incentives are over-used.

  Next Steps:

- Fix the January 25th Payment API bottleneck immediately.
- Integrate the Random Forest model into the CRM for real-time daily risk scoring.
- Launch the 15% discount experiment for the Medium Risk segment.